Week 09: October 26-30
Chemistry in Biology — The Building Blocks of Life

Subject/:الموضوع Biology **Unit Title: عنوان الوحدة Ecology** **Gradeالصف : 9**

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies Formative / Summative / Differentiation	Resources and links for Parents
Lesson 1:	**SUMMATIVE TEST** I can demonstrate my understanding of the concepts we have learned by completing the Criterion A Summative Assessment: • MUST — Levels 1–2: I can recall and identify key biological facts, terms, structures, and processes with some accuracy. • SHOULD — Levels 3–4: I can describe and explain biological concepts and use relevant evidence or examples to show my understanding. • SHOULD — Levels 5–6: I can apply and analyze my biological knowledge to unfamiliar situations, interpret information, and explain relationships using appropriate scientific terminology. • COULD — Levels 7–8: I can apply, analyze, and evaluate biological concepts accurately and independently, making well-supported conclusions and demonstrating a sophisticated understanding of unfamiliar problems.	8.1 Make goals clear, visible, and supported by short-term objectives. 7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models).	Study guide to be shared n Toddle and Teams

Lesson 2 Session 1: Carbohydrates & Lipids	I can compare carbohydrates and lipids by linking their building blocks and properties to their biological functions.	8.1 Make goals clear, visible, and supported by short-term objectives. 7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models).	Reading Essentials- The Building Blocks of Life.pdf
Lesson 3 Session 2: Proteins	I can explain protein functions by connecting amino acid sequences, folding and shape to biological roles.	8.1 Make goals clear, visible, and supported by short-term objectives. 7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models).	Reading Essentials- The Building Blocks of Life.pdf
Lesson 4 Session 3: Nucleic Acids	I can explain the roles of DNA and RNA by modelling nucleotide structure and tracing how genetic information is used.	8.1 Make goals clear, visible, and supported by short-term objectives. 7.2 Make learning relevant, authentic, and connected to students' backgrounds. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models).	Reading Essentials- The Building Blocks of Life.pdf
Homework **الواجبات**			
Teacher's notes/ Reminders **ملاحظات المعلم والتذكير**			

#الأسبوع Week # 09

Technology Integration دمج التكنولوجيا	Session 2 — Carbohydrates & Lipids: projected PPT, paper circles or digital sketch, Reading Essentials pp. 65–67. Session 3 — Proteins: projected PPT, paper strips/linking cubes/beads, folded paper or spring analogy, Reading Essentials pp. 67–68. Session 4 — Nucleic Acids: projected PPT, three paper shapes or digital drawing, nucleotide/arrow cards, Reading Essentials p. 68. Technology: PowerPoint, Teams/Toddle for sharing materials and collecting exit tickets.
Cross-curricular links الروابط بين المواد الدراسية	Session 2 — Chemistry: carbon bonding and molecular structure; UAE context: date fruit and waxy date-palm leaves; environmental science: reducing water loss in dry conditions. Session 3 — Physical Education: muscle function during a school sports event; chemistry: covalent peptide bonds and hydrogen bonds; UAE context: a runner training in Abu Dhabi. Session 4 — UAE context: date-palm seed growth; information and communication: instructions, messages, and models; chemistry: nucleotide components and chain formation.
External Assessment Integration دمج التقييمات الخارجية	NA
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	ATL Thinking — compare macromolecules and connect structure to function. ATL Communication — explain a model using accurate scientific vocabulary and evidence. ATL Collaboration / Social skills — build and discuss models with a partner.
Standards المعايير	HS-LS1-6: Construct and revise an explanation based on evidence for how carbon, hydrogen, and oxygen from sugar molecules may combine with other elements to form amino acids and/or other large carbon-based molecules. Biology: biological macromolecules and their functions, based on Reading Essentials, The Building Blocks of Life, pp. 65–68. Structure–function connection: molecular building blocks and properties relate to biological roles.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How do the structures of biological molecules help explain the roles they perform in living organisms?
Success Criteria معايير النجاح **(Measurable)**	Session 1 — Summative Session 2 — Carbohydrates & Lipids: MUST — DOK 1: Identify carbohydrate and lipid examples and state one function of each. SHOULD — DOK 2: Explain how carbohydrate structure and lipid properties relate to energy storage, structural support, or waterproofing. COULD — DOK 3: Use evidence from a UAE-themed scenario to recommend which macromolecule role matters most and justify the choice. Session 3 — Proteins: MUST — DOK 1: Identify amino acids as protein building blocks and peptide bonds as links between them. SHOULD — DOK 2: Explain how amino acid combinations and protein folding relate to structure and function. COULD — DOK 3: Use evidence to predict how a change affecting protein shape could influence a biological process in a UAE-themed case. Session 4 — Nucleic Acids: MUST — DOK 1: Identify the three parts of a nucleotide and name DNA and RNA.

	SHOULD — DOK 2: Explain how nucleotides form nucleic acids and distinguish the roles of DNA and RNA. COULD — DOK 3: Use a model to explain how genetic information links to protein production in a UAE-themed organism scenario.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Citizenship — Conservation; Values — Respect and responsibility; Culture — Heritage and connection to local environments.
	Area: environmental stewardship, scientific literacy, and appreciation of UAE biodiversity and heritage.
	Implementation: Session 1: Students use the UAE date palm as a context to connect carbohydrate-rich fruit with energy and waxy leaves with reduced water loss. Session 2: Students consider a runner training in Abu Dhabi and connect protein roles—including structural support and speeding reactions—to organism function. Session 3: Students use a date-palm seedling as a context for explaining how DNA stores instructions and RNA uses information to help make proteins. UAE examples are contexts for applying the biology, not claims about specific genes.